

Techniques for mastering complex tasks with AI

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Breaking down complex tasks into manageable steps can be a challenge when working with large language models (LLMs). Fortunately, specialized techniques can help you guide the AI through more complicated work. Two powerful approaches are **chain-of-thought prompting** and **prompt chaining**.

Chain-of-thought prompting

Chain-of-thought prompting is a prompting technique that involves requesting an LLM to explain its reasoning process step-by-step, from input to final output. This technique essentially asks the AI to "show its work," making its response more transparent and structured.

To use chain-of-thought prompting, simply include key phrases in your prompt, like:

- "Explain your reasoning"
- "Go step by step"

These additions indicate that the AI needs to trace its thought process, which often leads to more informative and accurate output.

Let's go through an example when this technique can be useful:

Consider an HR manager who is developing onboarding materials for a specific department. There's how chain-of-thought prompting could be used to identify actionable steps to handle the task.

Prompt:

Create a bulleted list outlining the major duties and responsibilities of a new entry-level design hire at an ad agency. Explain your reasoning step-by-step.

Notice the additional instructions "explain your reasoning step-by-step?" You can try using this prompt with and without this phrase to see the difference in AI's response.

By asking the AI to break down the logic and reasoning behind the duties it suggests, the information you get from the output will also have a rationale for why it was suggested. In this case, it may help the HR manager understand each **identify potential gaps** and **make informed decisions** about how to improve their current onboarding process.

Prompt chaining

While chain-of-thought prompting focuses on the reasoning within a single prompt, **prompt chaining** helps you tackle large projects by breaking them into a series of smaller, connected steps that are all in the same chat. It works like a factory assembly line: the output from one prompt is used as the input for the next, linking all your tasks together like a chain.



This technique involves three key steps:

1. **Task analysis:** Start by breaking down your complex task into a series of smaller, logical steps.
2. **Initial prompting:** Craft a focused prompt that asks the AI to complete just the first step.
3. **Input/output flow:** Use the output from the first prompt as the context for the second prompt. Continue this iterative flow until you complete the task.

Let's take a look at an example of a prompt chain:

Consider that you are planning a short vacation to Paris. You want to make sure you create an itinerary of the things you best most in a vacation, while maximizing logistics to be as efficient as possible with your time.

Prompt 1: I'm going to Italy (rome) for 3 days. I like art, historical sites, and parks. Suggest a few well-known places I could visit on my trip.

Prompt 2 (chained from Prompt 1): Using these locations, create a logical, day-by-day itinerary that minimizes travel time.

Prompt 3 (chained from Prompt 2): For each day of the itinerary, suggest a few restaurants located near each of the suggested locations.

By breaking the task into logical, digestible steps and using the output of one prompt as the specific input for the next, you transform the AI from a simple answer generator into a structured collaborator.

Combining prompt chaining with chain-of-thought

You can combine prompt chaining with chain-of-thought prompting to enhance the quality and accuracy of the problem-solving process at any stage.

Let's take a look at an example where we combine chain-of-thought prompting and prompt chaining:

Imagine you are heading your next book club meeting and you want suggestions for what book to recommend. You would like to recommend a fantasy novel, but fantasy is not a genre your book club typically reads. The fantasy book you select needs to be engaging without being overwhelming or too complex.

Prompt 1: I am heading a book club and would like fantasy book recommendations for people that are new to reading this genre. Suggest a few books that we could use.

Prompt 2 (using prompt chaining and chain-of-thought prompting): From that list, can you suggest which book you would recommend if we are looking for a fast-paced read? **Explain your reasoning**

This iterative process continues until a final book is selected and vetted. By breaking down the complex task with prompt chaining and incorporating chain-of-thought, you can address more difficult problems while ensuring more accurate results.

Limitations and best practices

While prompt chaining can be a useful technique, it's important to be aware of potential challenges that can occur—particularly with longer chains.

AI tools can struggle to remember context from earlier parts of the conversation as the prompt chain grows longer. This can lead to:

- Inconsistent responses
- Overlooking important details from earlier prompts
- Difficulty maintaining the overall objective of the task

To overcome these challenges and ensure effective prompt chaining, consider the following strategies:

- **Use checkpoints:** Periodically ask the AI to provide a brief summary of the overall goal.
- **Work on sub-tasks:** Divide very complex tasks into even smaller sub-tasks so that you can treat each as its own shorter chain before moving on to the next.
- **Recap and redirect:** If you notice an AI tool is deviating from the original goal, provide a recap of the essential information and redirect it back to the main objective.

By employing these strategies, you can generate more accurate and relevant output from an AI tool. As with any other prompting technique, prompt chaining relies on finding the right balance between providing necessary context and avoiding information overload.

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